

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni	Power	Omni	Power	Master

Section 6

Refrigerator system

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1. INTRODUCTION

The refrigerator system is used to reduce the helium boil-off substantially

This section describes the installation of a cold head SC-R8 with an HC-8E1 compressor (delivered with an F2000 type magnet).

The compressor needs watercooling (open loop or closed loop). For cooling requirements refer to the Planning Reference Book.

All components from the refrigerator system are charged with helium gas 4.5 (99,995%) (purity). Under a pressure of about 270 PSI.

2. INSTALLATION

2.1. COMPRESSOR INSTALLATION

2.1.1. UNPACKING

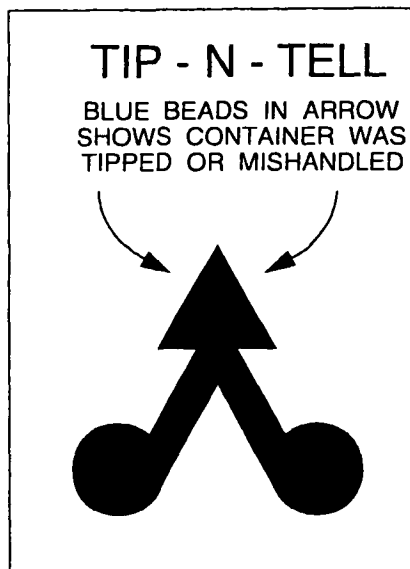
Open the shipping container, loosen the shipping restraints, and remove it's contents. The container also includes the compressor technical manual and the installation kit.

2.1.2. COMPRESSOR CHECKOUT AND CONNECTING TO POWER SUPPLY

1. Before powering up the compressor, check if the supply voltage to the compressor is correct. Check that the voltage is 220V +/- 10% between any two phases.
2. Be sure the power switch on the compressor is off.
3. Using 1/2" outer diameter tubing, connect the water cool supply and return lines to the fittings at the front of the compressor. Other couplings can also be used. Connect the other ends of both the supply and return lines to the cooling system of the hospital or to the waterchiller (if it was part of our delivery). Open the valve and adjust the flow to > 6 [l/min]. If no flow meter is present check the flow using a watch and a bucket.
4. Allow the compressor to reach room temperature. Read the pressure gauges. At room temperature the pressure should be between **273 and 283 psi**. Lower readings indicate that some of the gas charge has been lost.
5. Check the Tip-N-Tell indicator mounted on the side cover of the compressor. See Figure 1. If the sensor shows that the compressor has been transported in a forbidden position (up side down or very oblique) proceed as follows:
If the equalization pressure is between 273 and 283 psi continue the installation of the compressor. Run the compressor for 2 hours. The normal running time to check the compressor is 10 minutes. If the compressor shuts down during the two-hour test, replace the complete compressor. If the compressors runs without problem during this two hours continue with the installation and only write a problem report. to inform the PMG that a compressor has been arrived up side down.

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Figure 1 - Tip-N-TELL sensor



PV04 DRW

6. At this point the helium gas lines are not yet connected to the compressor. For an HC-8E1 compressor set the stand-alone operation switch to the 'Stand-Alone Operation' position. Switch on the hospital mains power. Activate the mains breaker of the compressor.

CAUTION

If the compressor starts but runs rough without building up significant pressure, turn it off immediately. It could be running in reverse. Change any of two phase on the transformer and switch on again.

When the compressor stabilizes, read the supply pressure gauge. The supply pressure should match the operation supply pressure: **310-340 psi**

Run the compressor for ten (10) minutes, then stop it.

7. Switch the stand-alone operation switch to the 'Normal Operation' position.
8. Check if the 'compressor shutdown' cable has been installed. (AA to FMX11). This cable must always be installed!

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2.2. INSTALLING THE HELIUM GAS LINES

WARNING

- When handling pressurized gas lines and other pressurized equipment, always wear eye protection.
- Never apply heat to a pressurized line.

CAUTION

The helium gas is pulsed through the copper helium lines. These pulses make the copper lines vibrate. Vibrations in the magnetic field of the copper lines makes that the two copper pipes are electrically charged. If the copper lines touch each other or another conductor an electrical discharge will take place. This discharge may cause artifacts in the images, the so-called 'spike' artifact.

To prevent this:

- Connect the copper lines from the RF-interface plate (RF room entry) via the shortest possible route to the cold head. Make sure that the copper lines do not touch each other or any other conductive part. If required use insulation tubing around the copper lines.
- Make sure the copper lines are not resting on the false ceiling. This to prevent vibration induced on the false ceiling. Ceiling vibrations alone can cause 'spike' artifacts, or induce vibrations in other parts which in turn may also cause 'spike' artifacts.

CAUTION

- Do not crimp the gas hose lines. Subsequent attempts to bend the hose lines may damage them.
- Check the condition of the gasket seal on the male half of each aeroquip coupling. Be sure the sealing surface on both the male and female halves are clean before connecting.
- Keep the gas line connection straight when making or breaking a coupling connection. Leakage may occur due to the weight of the gas line or due to a sharp bend near the connection.

NOTE

- Retain the protective caps and plugs to recover the couplings when they are not in use. They protect the module from damage and prevent entry of contaminants.
- Always hold the stationary nut on the hose coupling with one wrench while tightening the moveable coupling nut with the other wrench.
- Arrange the system components so that the interconnecting gas lines will be protected from stress and traffic. Observe the minimum bend radius. (57 mm when using a tube bender, 457 mm without using a tube bender).

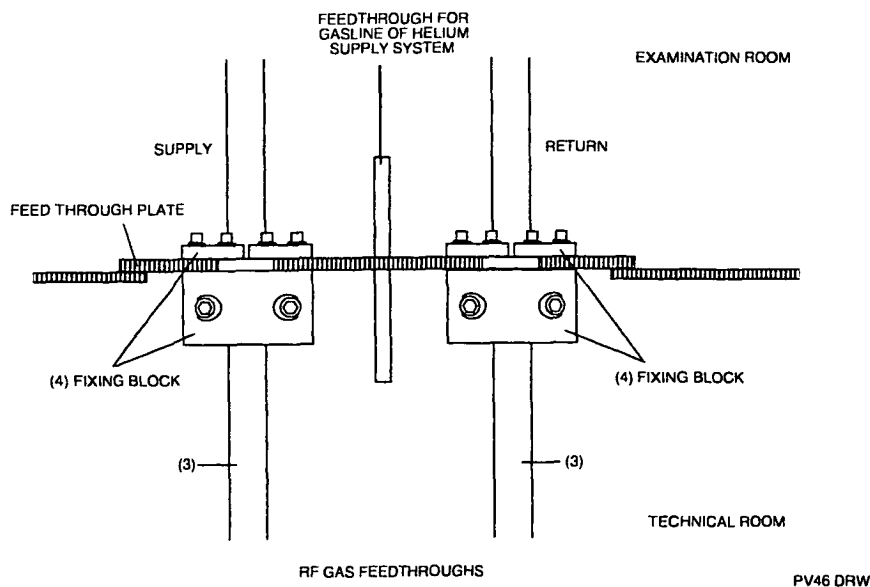
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Installation of the helium gas line feedthrough. (see Figure 2)

Procedure:

1. Install the gas lines from the technical room, through the opening of the RF gas feedthrough, to the examination room. Use, inside the examination room, the shortest possible route to the cold head.
2. Unscrew the two thin 'half moons' of the gas line connector.
3. Install the two thick 'half moons' around the gas line at the outside of the RF cages.
4. From the inside of the RF cage screw the two thin 'half moons' to the two thick 'half moons'.
5. Repeat this procedure for the other gas line.

Figure 2 - Gas line / RF cage connection

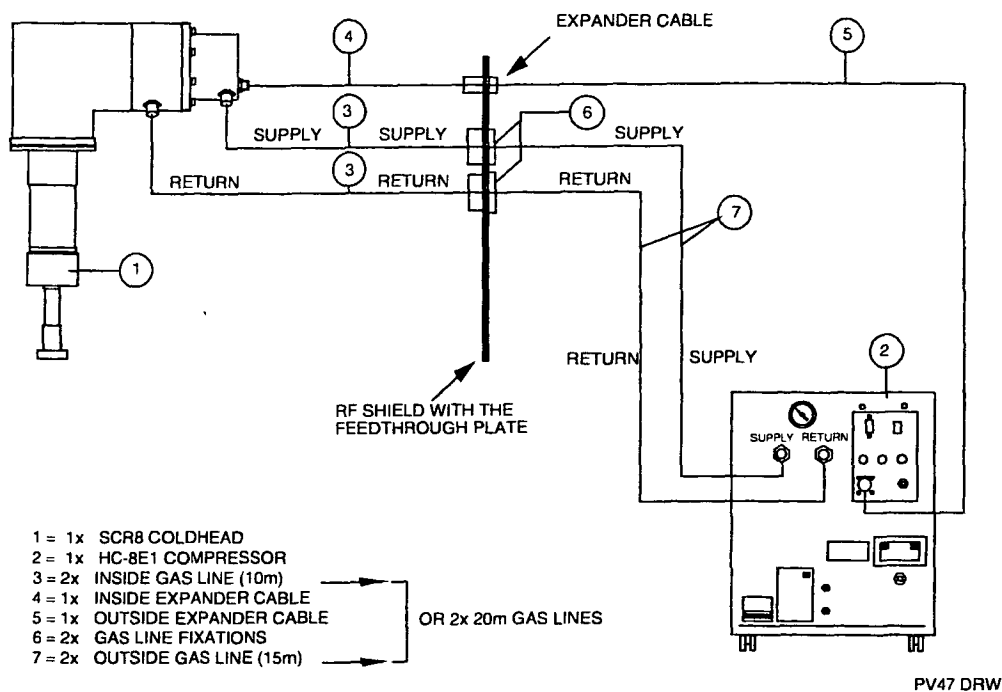


6. Stick a label **Supply** to both ends of a gas line.
7. Stick a label **Return** to both ends of the other gas line.
8. Remove the dust caps from the compressor supply and return gas couplings.
9. Remove the dust plugs from the couplings on one end of the supply and return gas line.
10. Connect the supply gas line to the supply coupling on the compressor. Use two wrenches to tighten the coupling. See Figure 3.
11. Connect the return gas hose line to the return coupling of the compressor.
12. Make sure that the copper lines do not touch each other or any other conductive part.
13. Install the insulation tubing around the copper lines.
14. Make sure the copper lines are not resting on the false ceiling.

Installation of the helium gas lines will be completed after the cold head is installed in the magnet.

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Figure 3 - Connecting the gas lines to the HC-8E1 compressor



2.3. COLD HEAD INSTALLATION

Tools

- Two open end wrenches 29mm and 31mm

Installation procedure

Preconditions:

- Compressor is switched off
- Mains cable is connected

1. Connect the helium gas lines to the cold head.
 - Connect the helium gas return line to the return connection of the cold head.
 - Connect the helium gas supply line to the supply connection of the cold head.
 - To be sure that disconnecting the gas lines is possible without using too much force, unscrew both lines 1/4 turn.
2. Attach the cold head power cable to the cold head.
3. Set the compressor to 'NORMAL' operation mode.
4. Switch on the compressor.
5. Check the gas lines when the refrigerator system is on. Because vibrations can damages the gas line reroute and/or support a part of the gas lines if needed.

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2.4. CHECKING THE BOIL OFF

Preconditions:

- Cold head should be running for more than 5 days.
- During the measuring period the cold head system should run 24 hours every day.
- No rampings during the measuring period.
- No helium refills during the measuring period.

Procedure:

1. Record the time, helium level and compressor hour counter.
2. After a period of minimal 4 weeks record the time, helium level and compressor hour counter again
3. Calculate the hours between the two readings.
4. Check with the compressor hour counter if the cold head system has been working during the entire period.
5. Calculate the helium boil off.

Specification:

F2000 1.5T: Maximum 0.065 litre / hour
 F2000 1T: Maximum 0.065 litre / hour
 F2000 0.5T: Maximum 0.065 litre / hour

NOTE

During the lifetime of the system record the liquid helium level and compressor hour counter at least once a week!
